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# 第二章　函数的导数(Derivative)

## 2.1　导数的定义(The Definition of Derivative)

### Multiple Choice Questions

C

1. Which of the following values is the average rate of change of $f(x)=\sqrt{x+2}$ over the interval $[-1,2]$?

(A) $-4$　　(B) $-\frac{1}{3}$　　(C) $\frac{1}{3}$　　(D) 3

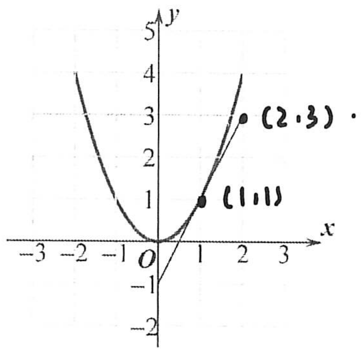
C

2. Let $f(x)=4-3x$. Which of the following is equal to $f'(0)$?

(A) 4　　(B) $-4$　　(C) $-3$　　(D) 4

D

3. The graph of $f$ and its tangent line at $x=1$ is shown below, find $f'(1)=($　　$)$.




(A) 0　　(B) $\frac{1}{2}$　　(C) 1　　(D) 2

A

4. Let $f(x)=kx^3-2x,(k\neq 0)$. The average rate of change of $f(x)$ over the interval $[-2,-1]$ is 3, find $k$.

(A) $\frac{5}{7}$　　(B) $\frac{1}{7}$　　(C) $-\frac{1}{7}$　　(D) $-\frac{5}{7}$

C

5. **Caculator** Let $y=\tan e^x-x^2+1$. Find $\frac{dy}{dx}\Big|_{x=0.5}=$

(A) $-12.056$　　(B) 0　　(C) 271.066　　(D) 273.066

C

6. **Caculator** The graph of $y=\sqrt{x^2+1}-3x$ crosses the $x$-axis at one point in the interval $[0,1]$. What is the slope of the graph at this point?

(A) 0.354　　(B) $-0.001$　　(C) $-2.666$　　(D) 0

$\lim\limits_{\Delta t \to 0} \frac{\quad}{\Delta t}$



B

7. The population of insects at time $t$ days is modeled by the function $P$ with first derivative $P'(t) = 0.1t^2 + 10t + 120$. Which of the following statements is true?

(A) At time $t = 10$, the population of insects is 230.

(B) At time $t = 10$, the population of insects is increasing at a rate of 230 insects per day.

(C) At time $t = 10$, the population of insects is decreasing at a rate of 230 insects per day.

(D) At time $t = 10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

$4 = k + b \qquad 9 = 3k$

8. If the line tangent to the graph of a function g at the point (1, 4) passes through the point $(-2, -5)$, what is the slope of this line?

$-5 = -2k + b$

A

(A) 3 (B) $-3$ (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

**Free Response Questions**

9. Let $f(x) = x^2 - 2x$

(a) Find the average rate of change of $f$ over $[1, 3]$.

(b) Find $\frac{f(2+h) - f(2)}{h}$.

(c) Find the instantaneous rate of change of $f$ at $x = 2$.

a

$\Delta y = 9 - 2\times 3 - (1-2)$

$= 9 - 6 - 1 + 2$

$= 4$

$\frac{4}{2} = 2$

b. $\frac{f(2+h) - f(2)}{h}$

$= \frac{(2+h)^2 - 2(2+h) - 2^2 + 4}{h}$

$= \frac{4 + h^2 + 4h - 4 - 4h}{h}$

$= h$

C $\lim\limits_{x \to 0} f(x) = \lim\limits_{x \to 0} (x - 2x)$

$= \lim\limits_{\Delta x \to 0} \frac{f(2+\Delta x) - f(2)}{\Delta x}$

$\lim\limits_{\Delta x \to 0} \frac{4 + \Delta x^2 + 4\Delta x - 4 - 2\Delta x - 2^2 + 4}{\Delta x}$

$\lim\limits_{\Delta x \to 0} \frac{\Delta x^2 + 2\Delta x}{\Delta x} \quad \Delta x + 2$

$= 2.$

10. Use the definition of derivative to find the derivative of $f(x) = x^3$, then find $f'(2)$.

$\lim\limits_{\Delta x \to 0} \frac{f(\Delta x + x_0) - f(x_0)}{\Delta x} = \lim\limits_{\Delta x \to 0} \frac{(\Delta x + x_0)^3 - (x_0)^3}{\Delta x}$

$= \lim\limits_{\Delta x \to 0} \frac{(\Delta x + x_0)^2(\Delta x + x_0) - (x_0)^3}{\Delta x}$

$= 3x_0^2$

$\therefore f'(2) = 3 \times 4 = 12$